QMSS 301: Quantitative Social Science Analysis and Big Data

Lab Assignment – Geospatial Analysis

Due on September 20, 2023, 11:59pm

We will be working with the *'Neighborhoods.shp'* data, the *'crime_by_neighborhood1.csv'* and *'Grocery_Stores_in_City_of_Detroit_Public_View.shp*' data for this assignment.
Read these datasets using appropriate functions and name them *'neighborhoods_data'*, *'crime_by_neigh',* and *'grocery_stores'* respectively.

1. A) Use a function to define what type of object these data files belong to. Write a one-line statement for your answers.
   B) Use a function to find out the CRS values for the data files. Note down your observations.
   C) How can you check if your data files have a projected or unprojected CRS? Interpret your answers.

2. Is the *'grocery_stores'* shapefile object projected? Write a function to appropriately project it if needed and name your projected variable *'grocery_stores_proj'*. Use a function to check for projection again.

3. Create an inner join between *'neighborhoods_data'* and *'crime_by_neigh'* data. Call your new data – *'combined_crime_neigh'*.
   Hint: You might want to check for the same column names before making the join. If not, rename one of the columns to match the other column.

4. We want to modify our combined dataset now so that is easily understandable. You can replace the dataset created above (question 3) with the modified one.
   A) Display only columns *OBJECTID*, *nhood_num*, *neighborhood*, *no_of_crime_inci*
   B) Change these column names to *NeighborhoodID*, *Neighborhood_Number*, *Neighborhood_Name*, *Crime_Inci_Number* respectively.
   Note: The geometry column can still show in the final dataset.

5. Using the modified data (created in question 4), find out which neighborhoods have the highest and lowest crime incidents. (Store in two new variables – *highest*, *lowest*)
   Hint: You can arrange them by the number of crime incidents and then combine with head/tail functions based on your preference.

6. Neighborhoods with crime incidents above 2500 are considered 'Unsafe' and those below are 'Safe'.

A) Create a new column in modified data called *'Neigh_Type'* that will store this information.
B) Finally, count the number of Unsafe and Safe neighborhoods using the *'Neigh_Type'* column. Give the name *'Neigh_Count'* to the variable storing this count. Store all of this information in a new variable called *'safety_info'*.
   Hint: Grouping and summarising will be helpful here.

Please submit your .r format file (in RStudio -> Click “Save as” -> “R Script”) via Canvas by the deadline.